Functional Annotation Report: pdxB (Q88L20 / PP_2117)

Erythronate-4-phosphate dehydrogenase — *Pseudomonas putida* KT2440 (PSEPK)

1. Summary (Answer to the Research Question)

pdxB encodes **erythronate-4-phosphate dehydrogenase (PdxB; EC 1.1.1.290)**, a cytoplasmic, NAD^+-dependent oxidoreductase of the **D-isomer-specific 2-hydroxyacid dehydrogenase superfamily**. Its primary function is to catalyze the **second committed step of the de novo (DXP-dependent, PdxA/PdxJ) pyridoxal 5′-phosphate (vitamin B6) biosynthetic pathway**: the NAD^+-dependent oxidation of the C2 hydroxyl of **4-phospho-D-erythronate** to yield **(R)-3-hydroxy-2-oxo-4-phosphonooxybutanoate** (2-oxo-3-hydroxy-4-phosphobutanoate). The enzyme acts as a soluble **homodimer in the cytosol** and is essential for vitamin B6 prototrophy; loss of pdxB produces a B6 auxotroph rescuable only by exogenous B6 vitamers via salvage.

Gene-identity verification: the UniProt description (erythronate-4-phosphate dehydrogenase, EC 1.1.1.290, gene *pdxB*, locus PP_2117), the InterPro domain set (IPR006139/IPR006140/IPR020921/IPR024531/IPR029753), and the D-2-hydroxyacid dehydrogenase family assignment all agree with the primary literature on PdxB, including the structurally characterized ortholog from the same genus (*Pseudomonas aeruginosa*). The identity is **confirmed and unambiguous**.

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2. Molecular Function — the Catalyzed Reaction

Reaction (EC 1.1.1.290):

> 4-phospho-D-erythronate + NAD^+ → (R)-3-hydroxy-2-oxo-4-phosphonooxybutanoate + NADH + H^+

- **Enzyme class:** oxidoreductase acting on the CH-OH group of a donor with NAD^+ as acceptor.
- **Cofactor:** NAD^+ (bound in a Rossmann-type nucleotide-binding domain).
- **Substrate specificity / stereochemistry:** the enzyme is a **D-isomer-specific 2-hydroxyacid dehydrogenase**, i.e., it stereospecifically oxidizes the 2-hydroxyacid substrate (4-phospho-D-erythronate) at C2. It is homologous to and mechanistically analogous to **SerA (D-3-phosphoglycerate dehydrogenase)**, the first enzyme of phosphorylated serine biosynthesis; the two share a common ancestor and analogous NAD^+/substrate/active-site residues [PMID 2681152].
- **Family evidence:** the crystallographically characterized *P. aeruginosa* ortholog is described as "a member of the d-isomer specific 2-hydroxyacid dehydrogenase superfamily" [PMID 17217963], and UniProt/InterPro assign the dedicated *Erythronate-4-P_DHase* signatures to Q88L20.

The reaction converts an alcohol (2-hydroxyacid) into a 2-keto acid, priming the carbon skeleton for the subsequent transamination step of the pathway.

Residue-level catalytic annotation (UniProt Q88L20, HAMAP-Rule MF_01825; residue identities verified against the sequence in this work). The catalytic active site comprises **Arg207, Glu236, and His253**, i.e., the **canonical His–Glu–Arg catalytic triad** of the D-isomer-specific 2-hydroxyacid dehydrogenase superfamily, with **His253 as the proton donor** (general acid/base). Mechanistically, His253 shuttles the substrate C2-hydroxyl proton during hydride transfer to NAD^+, Glu236 orients and tunes the pKa of the histidine, and Arg207 electrostatically stabilizes the substrate carboxylate/oxyanion transition state — the same logic used by SerA and other D-2-hydroxyacid dehydrogenases. **NAD^+** is coordinated by Asp146, Thr174, Ala205/Ser206/Arg207, Asp231 and Gly256; the **substrate** is anchored by Ser45, Thr66 and Tyr257. The formal pathway assignment is "pyridoxine 5′-phosphate from D-erythrose 4-phosphate: **step 2 of 5**" (UniPathway UPA00244). These positions are consistent with the *P. aeruginosa* structure and the 67% sequence identity established above.

Cross-database consensus. KEGG independently assigns PP_2117 to ortholog **K03473, "erythronate-4-phosphate dehydrogenase [EC:1.1.1.290]"**, in pathway **ppu00750 "Vitamin B6 metabolism"** (gene 2415325..2416467; 1143 bp / 380 aa), agreeing with the UniProt/HAMAP and BioCyc annotations.

3. Biological Process — the Pathway Context

PdxB operates in the **deoxyxylulose-5-phosphate (DXP)-dependent PdxA/PdxJ route of de novo pyridoxal 5′-phosphate (PLP / vitamin B6) biosynthesis**, the pathway characteristic of the γ-subdivision of proteobacteria (E. coli, *Pseudomonas*, *Photorhabdus*) [PMID 17217963]. The pathway assembles PLP from two branches that converge at PdxJ:

Branch feeding PdxB (the "4-carbon / hydroxythreonine" arm): 1. Erythrose-4-phosphate → 4-phospho-D-erythronate (erythrose-4-P dehydrogenase, Epd/GapB) 2. **4-phospho-D-erythronate → 2-oxo-3-hydroxy-4-phosphobutanoate — catalyzed by PdxB (this enzyme)** 3. → 4-phospho-hydroxy-L-threonine (transaminase SerC/PdxF, using L-glutamate) 4. → 3-amino-2-oxopropyl phosphate / (2-oxo-3-hydroxy-4-phosphobutanoate handling) by PdxA 5. **PdxJ** condenses the PdxA product with **1-deoxy-D-xylulose 5-phosphate (DXP)** to form **pyridoxine 5′-phosphate (PNP)** 6. **PdxH** oxidizes PNP → **pyridoxal 5′-phosphate (PLP)**, the active cofactor.

Thus PdxB provides the second enzymatic step of the erythronate-derived arm, generating the 2-keto acid that SerC then transaminates. This positions PdxB as a **committed, upstream enzyme of essential cofactor biosynthesis** rather than a peripheral or pleiotropic factor.

Essentiality / genetic evidence. In *Photorhabdus luminescens*, a transposon insertion in *pdxB* produced a strain "required for de novo vitamin B6 biosynthesis"; the mutant showed growth deficiency in nutrient-poor medium that was "restored … by supplementation with pyridoxal 5′-phosphate (PLP)" and by other B6 vitamers (pyridoxal, pyridoxine, pyridoxamine) via salvage [PMID 27060119]. In *E. coli*, "PdxB (erythronate 4-phosphate dehydrogenase) is expected to be required for synthesis of the essential cofactor pyridoxal 5′-phosphate (PLP)" [PMID 31712440]. Because *P. putida* KT2440 encodes *pdxB* (PP_2117), it uses this DXP-dependent pathway rather than the single-step DXP-independent PdxS/PdxT (ribose-5-phosphate + glutamine) route found in *Bacillus* and many other taxa.

Downstream significance. PLP is the obligatory cofactor for a large set of transaminases, decarboxylases, racemases, and related enzymes (amino-acid and one-carbon metabolism). PdxB activity therefore underpins broad PLP-dependent metabolism, but its *precise* role is the single oxidation step above — the wider metabolic effects of pdxB loss are secondary consequences of PLP depletion, not additional catalytic activities.

4. Subcellular Localization

- **Cytoplasmic / cytosolic.** UniProt Q88L20 explicitly annotates **SUBCELLULAR LOCATION: Cytoplasm** (HAMAP-Rule MF_01825), and **SUBUNIT: Homodimer**. PdxB is a soluble enzyme: the *P. aeruginosa* ortholog is a **homodimer of 380-residue subunits**, crystallized as a soluble protein with bound NAD, with no membrane association [PMID 17217963].
- UniProt/InterPro annotate no signal peptide, lipobox, or transmembrane segment for Q88L20, consistent with a location in the bacterial cytoplasm where its substrate (a phosphorylated central-metabolism intermediate) and cofactor (NAD^+) reside.

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5. Structural Basis (Evidence from the Genus-level Ortholog)

The first PdxB structure was solved for *Pseudomonas aeruginosa* D-erythronate-4-phosphate dehydrogenase in complex with NAD [PMID 17217963]:

- **Quaternary structure:** homodimer; the **C-terminal dimerization domain has a unique fold** largely responsible for dimer formation (matching InterPro IPR024531, "Erythronate-4-P_DHase_dimer", assigned to Q88L20).
- **Domain organization per subunit:** (i) a **lid domain**, (ii) a **nucleotide (NAD)-binding domain** (Rossmann-like; InterPro IPR006140 "D-isomer_DH_NAD-bd"), and (iii) the **C-terminal dimerization domain**.
- **Catalytic domain:** InterPro IPR006139 ("D-isomer_2_OHA_DH_cat_dom") and the conserved signature IPR029753 ("D-isomer_DH_CS") mark the catalytic 2-hydroxyacid dehydrogenase machinery (conserved Arg/Glu/His catalytic residues typical of the superfamily).
- **Conformational dynamics:** the two subunits in the crystal captured different ligand combinations and open vs. more-closed active-site clefts, illustrating the hinge-bending domain closure over the phosphorylated substrate that accompanies catalysis.

Because *P. putida* and *P. aeruginosa* are congeners with highly conserved PdxB sequences, this structure is a strong homology template for the localization, oligomeric state, cofactor usage, and catalytic mechanism of PP_2117.

Quantitative sequence support (this work). Global Needleman–Wunsch alignment of UniProt sequences confirms strong orthology: P. putida PdxB (Q88L20, **380 aa**) is **67.3% identical** to the crystallized *P. aeruginosa* PdxB (Q9I3W9, also **380 aa**; the enzyme of

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P 17217963
and **46.5% identical** to *E. coli* PdxB (P05459, 378 aa). Both *Pseudomonas* proteins are exactly the same length. A Rossmann-type nucleotide-binding fingerprint (**GAGEVG**) is present at residues ~123–128 of Q88L20, consistent with the NAD-binding domain. These identities (well above the ~30% "twilight zone") justify confident transfer of the P. aeruginosa structure and catalytic mechanism to PP_2117.

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6. Supported and Refuted Hypotheses

Supported: - PdxB is erythronate-4-phosphate dehydrogenase (EC 1.1.1.290), an NAD^+-dependent D-2-hydroxyacid dehydrogenase. ✔ (UniProt/InterPro +
P 17217963
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P 2681152
- It performs the second step of the DXP-dependent de novo PLP (vitamin B6) pathway and is essential for B6 prototrophy. ✔
(
P 27060119
P 31712440
)
P 17217963
- It is a soluble cytoplasmic homodimer with lid / NAD-binding / dimerization domains. ✔
(;
P 17217963
InterPro)

Refuted / excluded: - PdxB is **not** a transporter, structural protein, or signaling molecule; it is a soluble metabolic oxidoreductase. - *P. putida* does **not** rely solely on the DXP-independent PdxS/PdxT PLP route — the presence of pdxB places it in the DXP-dependent pathway. - No evidence supports a membrane-bound or secreted localization.

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7. Limitations and Future Directions

- **No PdxB study exists on P. putida KT2440 itself.** Functional/structural evidence is transferred from the same genus (*P. aeruginosa*) and from *E. coli*/*Photorhabdus* orthologs; direct in vitro kinetics (Km, kcat, substrate specificity) for PP_2117 have not been reported.
- **Kinetic promiscuity / underground metabolism.** In *E. coli*, PdxB deletion can be partially bypassed by promiscuous activities of other enzymes; whether *P. putida* has analogous bypasses is unknown (the Kim et al. 2019 study, P 31712440, addresses this in *E. coli*).
- Future work: recombinant expression and steady-state kinetics of PP_2117; confirmation of NAD^+ vs. $NADP^+$ preference; genetic essentiality/auxotrophy test in KT2440; an experimental (rather than homology-modeled) structure.

References

- P 17217963 — Ha et al. (2007). *Crystal structure of D-erythronate-4-phosphate dehydrogenase complexed with NAD.* (Structure, family, mechanism, homodimer/domains; *P. aeruginosa* ortholog.)
- P 2681152 — Schoenlein, Roa, Winkler (1989). *Divergent transcription of pdxB and homology between the pdxB and serA gene products in E. coli K-12.* (Gene, SerA homology, NAD^+/2-hydroxyacid dehydrogenase inference, pdxB-hisT operon.)
- P 27060119 — Sato, Yoshiga, Hasegawa (2016). *Involvement of Vitamin B6 Biosynthesis Pathways in the Insecticidal Activity of Photorhabdus luminescens.* (pdxB required for de novo B6; auxotrophy rescued by PLP/vitamers; salvage.)
- P 31712440 — Kim et al. (2019). *Hidden resources in the E. coli genome…* (PdxB required for PLP synthesis; underground/bypass metabolism.)

- **UniProt Q88L20** (HAMAP-Rule MF_01825) and **InterPro** IPR006139/IPR006140/IPR020921/IPR024531/IPR029753 (domain and family annotation for PP_2117).
